

## **Mapping in Antarctica**

C.S. BEWLI

Survey of India

### **Abstract**

The paper describes the various steps involved in the details and contour survey of an area of 1.4 sq km of Indian Station Maitri and around on scale 1:5,000 with contours at 5 m vertical interval carried out resulting in a map of the area.

In addition to this, survey facility was extended to National Physical Laboratory, Geological Survey of India and Indian Army Team for carrying out their scientific activities in an efficient manner.

### **Introduction**

Maps are a means of communication in graphical form and form the basis for development in every country. Previously, primary maps were mainly prepared for use by defence forces. But, after independence, with the rapid development of our country, the demand of maps has increased considerably and thus the number of map users has multiplied several times. Demand on land is increasing day by day. This made importance of maps on various scales greater. The appreciable development in the scientific activities carried out during the first nine Indian Scientific Expeditions to Antarctica necessitated the provision of geographical and geophysical control in the Indian Station Maitri and around, located in East Antarctica. Thus, Survey of India (SOI), the national survey and mapping organisation of our country was identified by the Department of Ocean Development (DOD) to carry out mapping task from tenth Indian Scientific Expedition to Antarctica.

The survey of an area around Indian Station Maitri on scale 1:5,000 with contour interval of 5 m will be of great assistance for planning and execution of scientific and engineering activities in a scientific manner. This need was felt since long.

Hence during XI Indian Expedition to Antarctica, an area of 1.5 sq km was covered by a team of three scientists of SOI who surveyed the area around

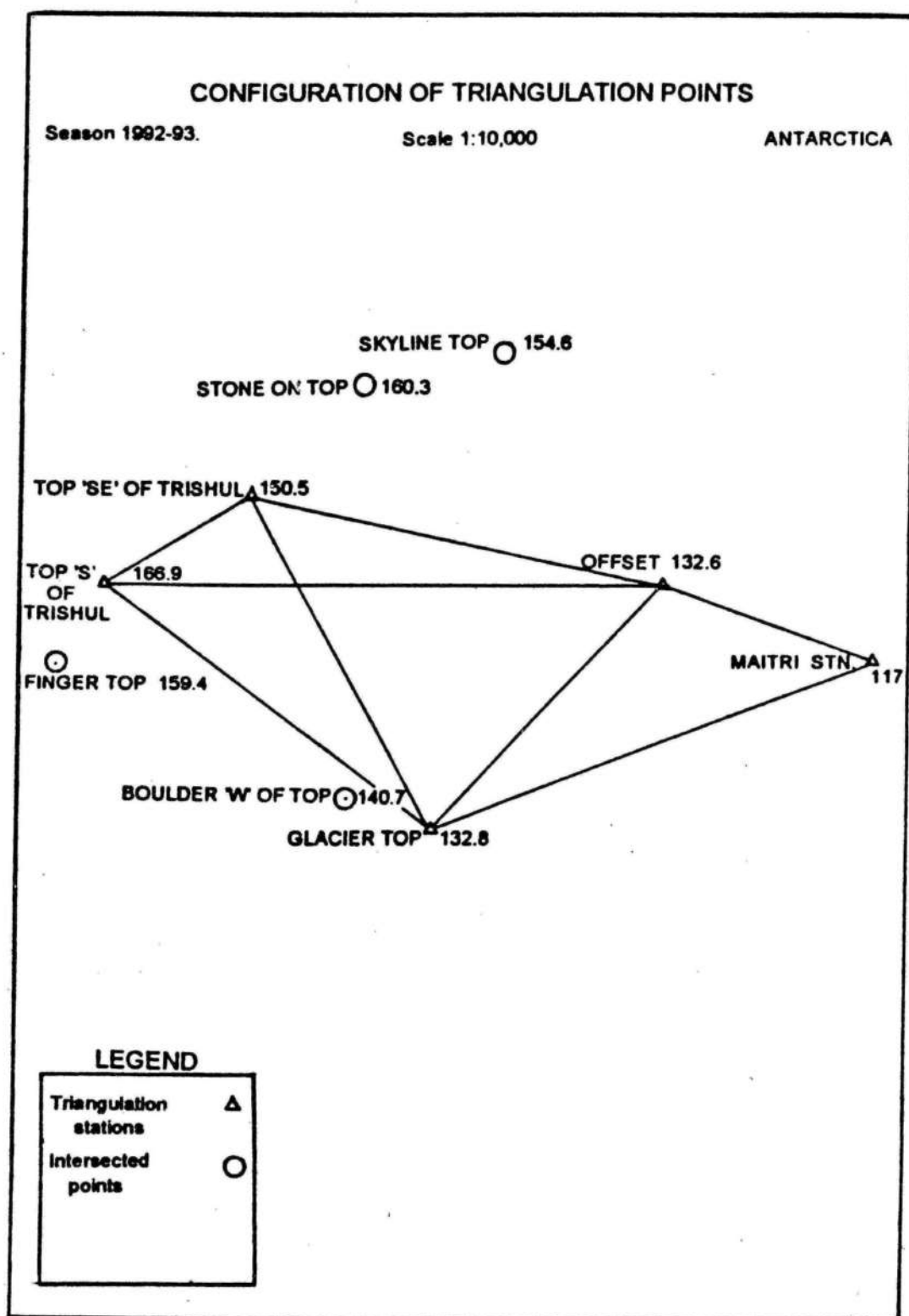


Fig. 1. Configuration of triangulation points

Maitri Station. This survey was extended by the Survey of India team during XII Antarctic Expedition.

### **Objectives**

1. To verify the earlier survey around Maitri on scale 1:5,000 with contour interval of 5 m as per standard departmental practice.
2. To extend ground control points on the western side of Maitri for further survey of an area adjoining the western edge of the survey carried out during XI Expedition.
3. To carry out details and contour survey on scale 1:5,000 with vertical interval of 5 m using the control points established in 2 above.

### **Verification**

The earlier ground survey carried out during XI Expedition on scale 1:5,000 with vertical interval of 5 m was verified and an Insertion & Deletion Guide was prepared, wherein the details existing earlier and no more on the ground now have been deleted and the details which have come up since the earlier survey have been incorporated using normal plane tabling techniques.

### **Reconnaissance**

A thorough reconnaissance was carried out for providing ground control points uniformly distributed in the area of work in such a way that any height could be computed with respect to them. Cairns were made for correct identification of the control points. The configuration of the control points is shown in Figs 1 & 2.

### **Triangulation**

Triangulation by using Wild T2 Theodolite was carried out establishing 4 ground stations and 4 intersected points covering an area of 1.4 sq km. One permanent ground control point, provided by Survey of India team during X Antarctic Expedition located at Maitri has been used as origin for this survey. To have good agreement of the survey the following observation procedures were followed:



*(i) For Ground Stations*

Horizontal angles to stations were observed on both the faces at three different zeroes. The agreement in the angles obtained independently at three zeroes was kept within 20 seconds. Vertical angles for deduction of heights were observed twice on both the faces. The agreement of vertical angles between the faces was kept within 30 seconds.

*(ii) For Intersected Points*

Horizontal angles were observed for intersected points on both the faces at two different zeroes and vertical angles on both the faces once.

**Computations**

The base length chosen to carry out triangulation was the extension of earlier survey, thus the length between Maitri Station and Offset was taken as Base.

Heights and coordinates of the ground control points and intersected points were computed with the observations made during triangulation.

**Details and Contour Survey**

The plane table section was divided in regular grids, each grid of size 2 cm × 2 cm (*i.e.* 100 m × 100 m). The ground control points were plotted on P.T. Section.

A total number of 35 fixings (plane table positions) were made to cover an area of 1.4 sq km. The height of the fixings were always obtained from ground stations/intersected points and not from other deduced heights, this has added to the accuracy of the contours. From these 35 fixings, a mesh of 320 heights was established. It was by and large ensured that minimum one height is available at every centimetre on paper in order to have better configuration of the contours.

**Fair Drawing**

After the field work, film negatives of this survey and the earlier survey were taken and mosaiced together. Impression of this mosaic was taken on a good quality drawing paper by photographic means. Drawing of the complete area was done all over again, the result is a map of the area (see Fig. 1 in reduced form).

### **Technical Assistance to Other Scientific Teams**

#### *(a) National Physical Laboratory (NPL)*

On request from NPL Team Leader, Dr. H. N. Dutta, ground survey on scale 1:1,000 was carried out to give relative positions of their complete establishment in Maitri. This survey will assist NPL Scientists (Delhi) to plan for their future scientific establishment keeping in mind the distance of the existing ones. Also, this is an essential requirement in computing the phase velocity of the various wave phenomena observed in the cold static Antarctic climate. The survey result is shown in Fig. 3.

#### *(b) Geological Survey of India (GSI)*

GSI Team Leader, Dr. S. Mukherji requested for the survey of Dakshin Gangotri Glacier Snout in order to monitor its movement. The work involved a number of intersections to fix the limits of snout and adjoining area. The survey was carried out with respect to an already existing GSI point. The survey of the snout is shown in Fig. 4. This has enabled GSI scientists to continue their study, they had been doing for the last six expeditions.

#### *(c) Army Team*

The Leader of the Army Team had requested to give the difference of height between the two points in order to lay the oil pipeline for the newly erected Generator Shed. The survey indicated the difference of 1.9 m between the two points which was found enough for the free flow of oil.

### **Conclusions**

An area of 1.4 sq km was covered. The planning of the whole day's work was made in advance in order to cover the area with optimum number of fixings and heights. The uniform distribution of the ground control points helped in this respect. After putting in our best efforts and utilising every bit of available time, undeterred by hostile weather, an area of one and a half times as scheduled was covered. It will not be out of place to mention that ground survey is an outdoor task which needs whole day trodding in difficult terrain of Schirmacher where the mapping work was to be done at sub-zero temperatures without use of gloves. After completion of the field work fair drawing of the area was done resulting in a map on scale 1:5,000 with 5 m contour interval which can be obtained from officer-in-charge, Map Record & Issue Office, Map Publication, Survey of India, Dehradun-248001 (Price Rs. 16/-).

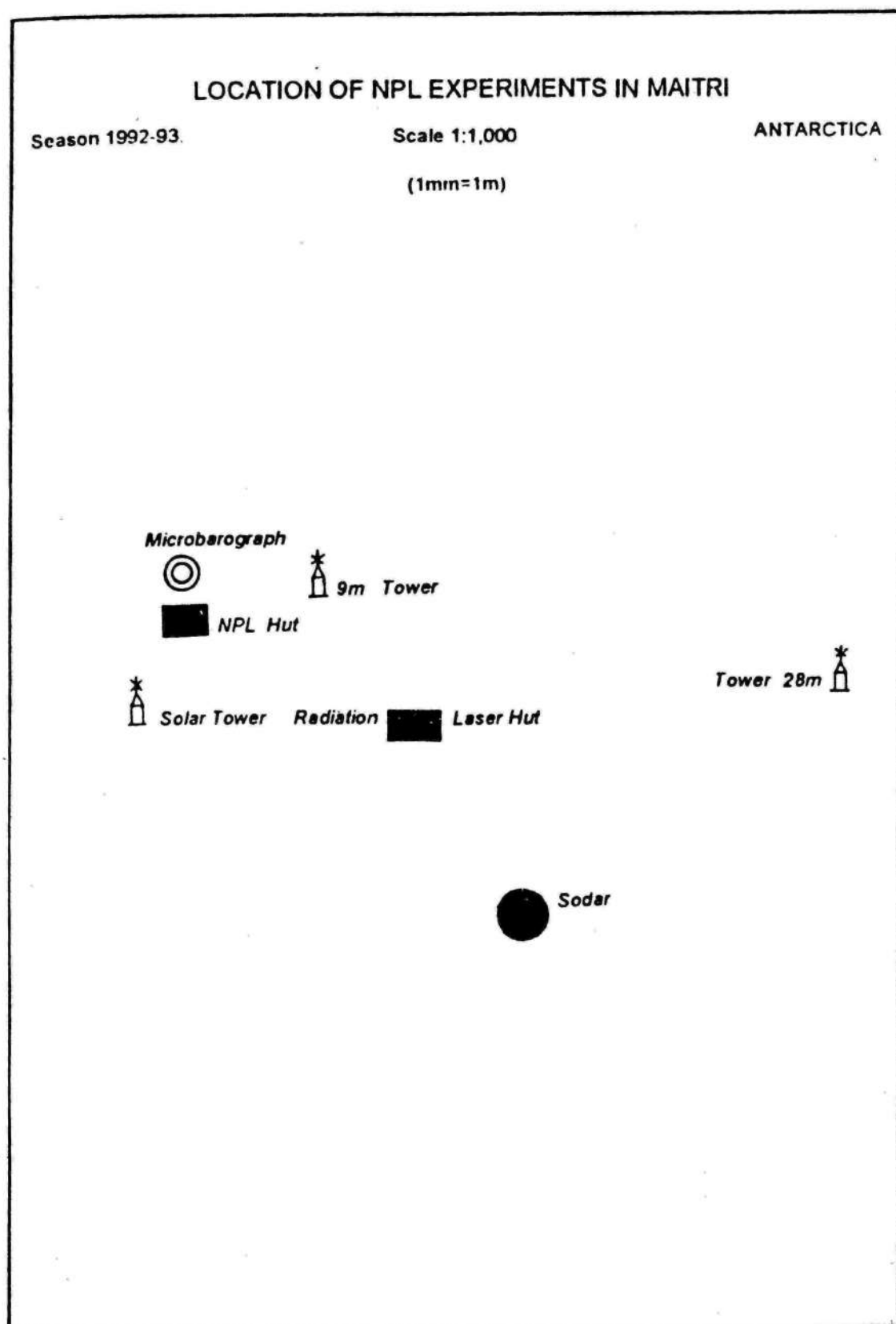
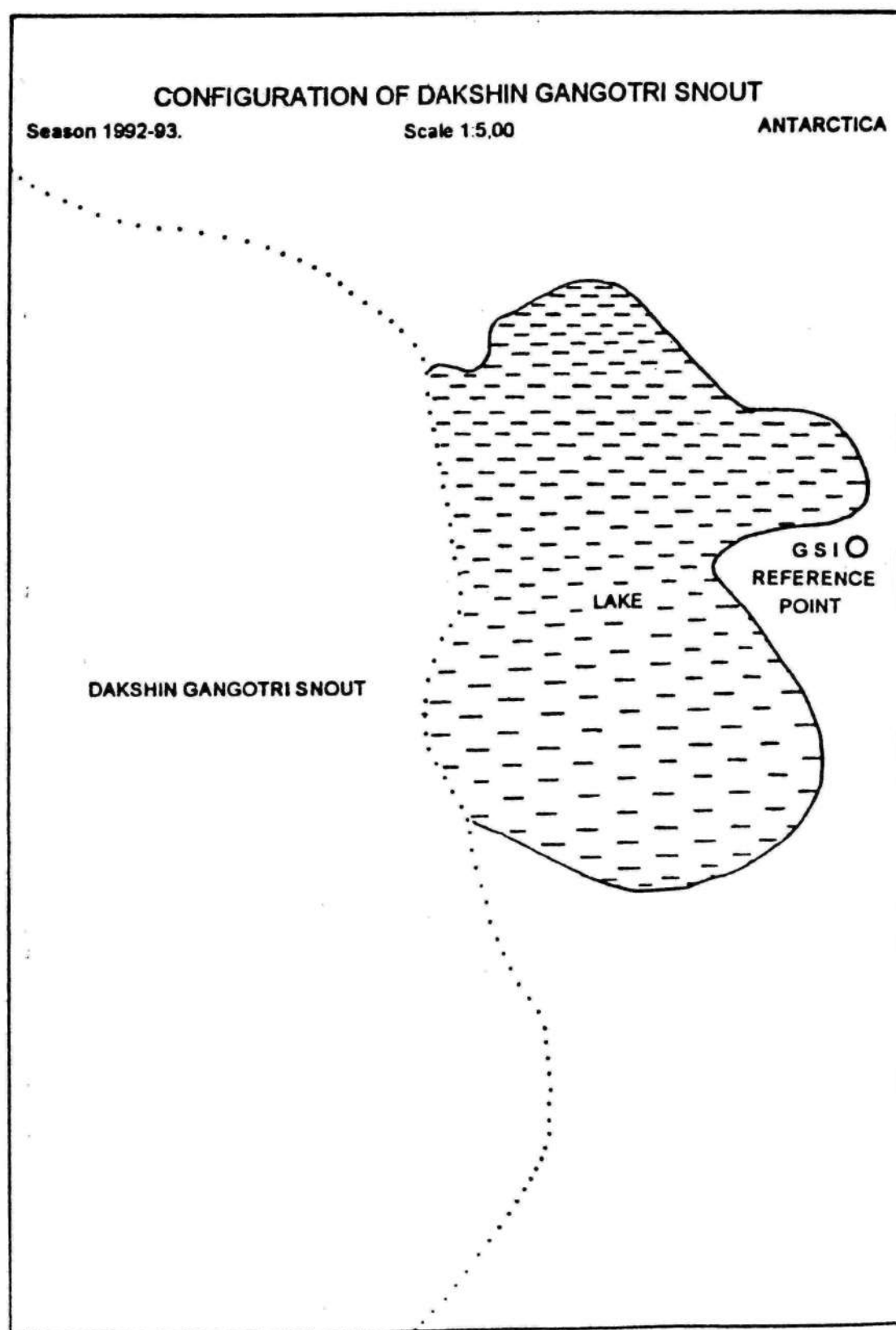


Fig. 3. Location of NPL experiment in Maitri



*Fig. 4. Configuration of Dakshin Gangotri snout*

### **Future Recommendations**

1. An aerial photography on scale 1:10,000 should be flown for the whole strip typed Schirmacher area (3 km × 16 km approx.) which consists of low undulating hills. For achieving this an aerial photography camera (gyro stabilised) should be fixed in MI-8 or any other helicopter, if need be assistance may be taken from 106 squadron of Indian Air Force which deals with aerial photography.
2. Ground control points should be established on the whole Schirmacher range in order to survey the area initially on scale 1:25,000 with 10 m vertical interval and later on in any larger scale as per the requirement.
3. Actual ground survey should be carried out on scale 1:25,000 with vertical interval at 10 m on photogrammetrically plotted air survey sections.